

Experiment 1: Representational Crowding as a function of scene density

Research Question: Does representational overlap between object tokens in a vision encoder scale with scene density, and at which encoder layers?

If crowding reflects a loss of discriminability between concurrently represented objects as visual clutter increases, then scenes containing more objects should show greater similarity among their object tokens, with this effect concentrated in the mid-to-late layers where crowding operates.

Setup: In this experiment, the mean pairwise cosine similarity was measured among all object tokens in each scene, computed layer-by-layer within the standalone SigLIP-SO400M Encoder. Scenes were grouped by object count (2, 4, 8, 16, and 32 objects), and similarity was averaged within each count group at each of the 27 encoder layers.

Results:

Finding 1: Object-token similarity follows a U-shaped trajectory across depth. Mean pairwise similarity rises steeply over the first several layers, from 0.775-0.800 at layer 0 (per condition: 0.792, 0.775, 0.787, 0.782, 0.800, 0.780 for 2, 4, 8, 16, 30, 32 objects) to a plateau of 0.905-0.922 (layer 7: 0.922, 0.908, 0.915, 0.907, 0.913, 0.905), and then declines steadily through the second half of the network. Early tokens thus place all object tokens in a relatively shared representational space, while late layers progressively differentiate them.

Finding 2: Similarity is count-ordered precisely where crowding is significant. The curves for different object counts separate in the mid-to-late layers (14-26, shaded region, Figure 1) And the ordering tracks scene density: at layer 18, scenes with 30 and 32 objects retain the highest pairwise similarity (0.867 for both), followed successively by 16 (0.851), 8 (0.832), 2 (0.797) and 4 (0.793),. The separation is maximal around layers 21-23, where similarity spans approximately 0.79-0.87 between the sparsest and densest conditions. Denser scenes therefore maintain greater representational overlap between object tokens throughout this layer range, consistent with increased confusion among concurrently represented objects as clutter increases.

Finding 3: A late-layer, count-independent representational shift. All count conditions exhibit a synchronized dip at layer 25 (spanning approximately 0.79-0.85 across conditions) followed by a rebound at layer 26. Because this dip-and-recovery pattern is invariant to object count, it reflects a property of the encoder's late-layer processing rather than a crowding effect, and we exclude it from interpretation of density-dependent effects

Limitations:

The numeric values reported here are read off the figure (to be added) rather than extracted from the underlying per-layer data; before submission, the exact values from the CSV should replace the approximations. Namely, the precise layer of maximal count separation and the trough values.

This experiment characterizes global pairwise similarity by scene object count; it does not analyze the similarity of specific queried pairs, and so it cannot speak directly to trial-level effects.

The count ordering is a property of the standalone SigLIP encoder; it should not be generalized to multimodal vision towers without the analogous results.

Because the count-ordered separation in the shaded region is described rather than statistically tested here, a formal test of the count x layer interaction (or pairwise count contrasts within layers 14-26) would strengthen the claim of monotonic density ordering.

num_objects	2	4	8	16	30	32
layer						
encoder.layers.0	0.791939	0.774572	0.787038	0.781745	0.800120	0.780312
encoder.layers.1	0.853230	0.835246	0.846433	0.838504	0.849324	0.833402
encoder.layers.2	0.888176	0.872470	0.882004	0.874009	0.882331	0.868775
encoder.layers.3	0.906148	0.892105	0.900007	0.891825	0.897872	0.886771
encoder.layers.4	0.905857	0.892058	0.900077	0.892802	0.898193	0.889265
encoder.layers.5	0.909788	0.895885	0.903385	0.896110	0.901556	0.893287
encoder.layers.6	0.918612	0.904658	0.911660	0.904161	0.909392	0.901749
encoder.layers.7	0.922224	0.908482	0.914928	0.907400	0.912576	0.905392
encoder.layers.8	0.918854	0.904479	0.911184	0.904568	0.911278	0.903810
encoder.layers.9	0.919965	0.904208	0.911761	0.905903	0.913547	0.906657
encoder.layers.10	0.921031	0.907480	0.916599	0.912755	0.921762	0.915159

encoder.layers.11	0.917713	0.902334	0.911619	0.909887	0.920560	0.914985
encoder.layers.12	0.910369	0.894261	0.904753	0.904418	0.915370	0.910840
encoder.layers.13	0.899819	0.885308	0.898851	0.901457	0.913810	0.909043
encoder.layers.14	0.865535	0.854832	0.875259	0.883570	0.900154	0.895890
encoder.layers.15	0.844844	0.836067	0.860437	0.872532	0.889843	0.887427
encoder.layers.16	0.820358	0.812942	0.842489	0.858951	0.875755	0.875098
encoder.layers.17	0.804723	0.800001	0.834333	0.854046	0.870046	0.870408
encoder.layers.18	0.797217	0.792811	0.831943	0.851463	0.867091	0.866730
encoder.layers.19	0.790060	0.788270	0.826878	0.846543	0.862295	0.861050
encoder.layers.20	0.791745	0.791660	0.828807	0.848162	0.863880	0.862054
encoder.layers.21	0.793845	0.793402	0.830364	0.850321	0.865496	0.864217
encoder.layers.22	0.810733	0.807987	0.844765	0.864469	0.876928	0.875223
encoder.layers.23	0.821725	0.819112	0.850960	0.867076	0.876772	0.874704
encoder.layers.24	0.829559	0.820665	0.846654	0.860729	0.871736	0.869161
encoder.layers.25	0.793543	0.785522	0.815599	0.833608	0.848256	0.844898
encoder.layers.26	0.795053	0.789180	0.828319	0.853006	0.877217	0.874116

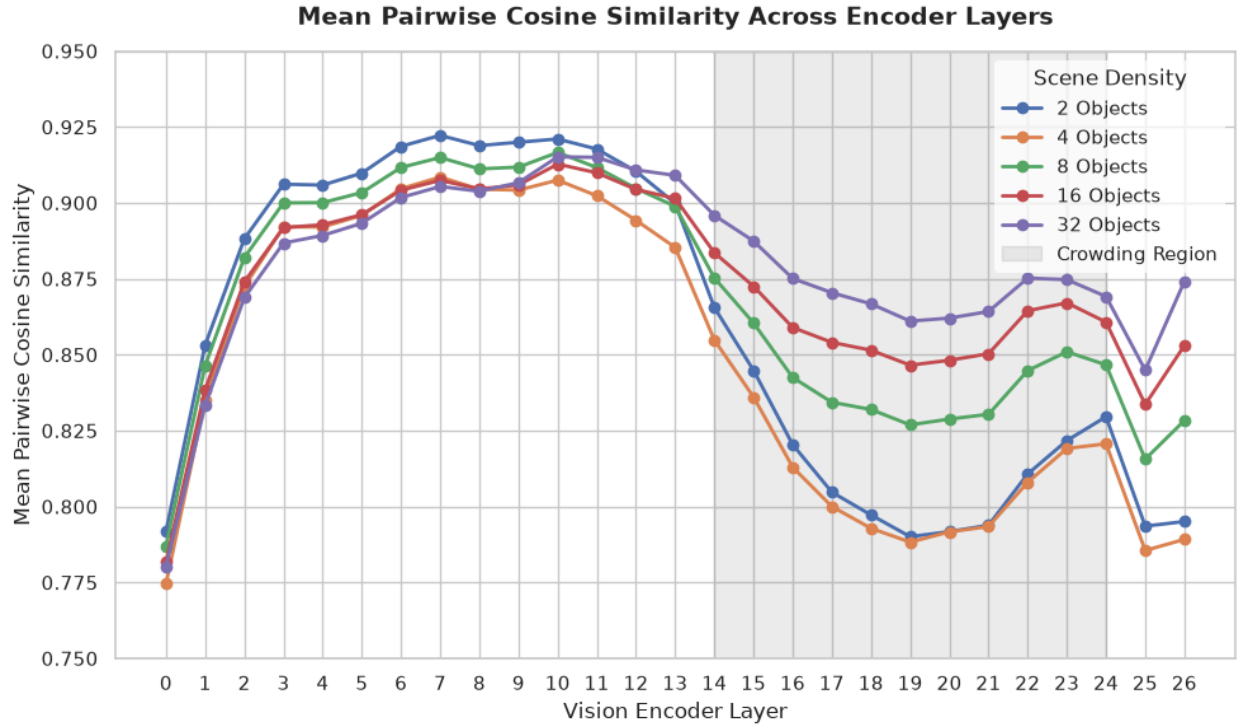


Figure 1. Mean pairwise cosine similarity between object tokens as a function of vision encoder layer (0-26), separately for scenes containing 2, 4, 6, 16, 30, and 32 objects. The shaded region marks the mid-to-late layers where similarity scales with scene density. All conditions show a synchronized dip at layer 25 and rebound around layer 26, which is excluded from crowding interpretation.